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Project: Hydro-Risk Price Forecasting Model

A Regime-Based Analysis of Hydrological Storage and Wholesale Electricity Price Risk in New Zealand

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Managerial Abstract

Hydrological uncertainty creates a persistent and recurring decision challenge for Wellington Electricity: how to recognise periods of escalating wholesale price risk early enough to support timely hedging decisions, without overreacting to short-term market noise. In hydro-dominated electricity systems, wholesale prices are strongly influenced by expectations about future water availability rather than contemporaneous conditions alone, implying that risk evolves in a state-dependent and forward-looking manner (Weron, 2014; Joskow, 2019).

Rather than attempting to forecast short-term prices, this project focuses on how wholesale price risk evolves across distinct hydrological states. National storage conditions are classified into interpretable regimes, and their dynamics are examined to understand how system stress develops and persists over time. Evidence from the New Zealand electricity system highlights that periods of low storage tend to be prolonged rather than transient, reinforcing the importance of monitoring risk escalation over planning horizons relevant for hedging decisions (Electricity Authority, 2023).

Wholesale price outcomes are evaluated conditionally on hydro-storage regimes, allowing risk to be characterised in distributional terms rather than through average price movements alone. This regime-based perspective provides a structured way to link observable system states to changes in the severity and persistence of potential cost outcomes, particularly under stress conditions. Scenario and simulation analysis are then used to express these regime-dependent risks over relevant planning horizons in exposure terms that inform budgeting discussions, support risk appetite framing, and enable structured review of hedging decisions.

The project's primary contribution is an operational decision support design that consolidates regime status, regime transition risk, and exposure measures into a coherent monitoring framework. By shifting attention away from point expectations and toward state dependent risk escalation, the framework supports more disciplined, transparent, and defensible hedging discussions under hydrological uncertainty.

Chapter 1: Introduction

1.1 Research Topic and organisational context

New Zealand's electricity system is characterised by a high reliance on hydro generation, with water storage acting as the primary intertemporal buffer between supply and demand. As a result, wholesale prices reflect not only current system conditions but also expectations about future water availability and the opportunity cost of stored energy (Joskow, 2008; Joskow, 2019).

During periods of sustained low inflows or declining storage levels, reduced hydro flexibility increases reliance on higher-cost marginal generation, raising the likelihood of scarcity pricing and extreme price outcomes. These features are well documented in electricity market research and are particularly pronounced in systems where storage constraints limit short-run supply responses (Borenstein, 2002). In the New Zealand context, regulatory assessments consistently identify hydrological conditions as a central driver of wholesale market performance and price volatility (Electricity Authority, 2022).

Wellington Electricity operates as a regulated electricity distribution business with exposure to wholesale electricity prices through procurement and hedging arrangements. Although it is not a generator, its cost base and risk profile are directly affected by wholesale market volatility. These dynamics create an ongoing operational challenge for managing cost risk under hydrological uncertainty.

Against this backdrop, the project focuses on understanding how hydrological conditions translate into wholesale price risk and how this risk can be monitored and managed in a structured, data-driven manner. The analysis is framed explicitly around applied decision support rather than short-term price forecasting, reflecting the organisational need for defensible risk signals rather than point predictions.

1.2 Definition of the business decision problem

The central business decision problem addressed in this project is determining when and how Wellington Electricity should adjust wholesale risk positions, particularly hedging actions, in response to evolving hydrological conditions. Specifically, the challenge is to identify periods when hydrological stress meaningfully increases the probability of adverse price outcomes, while avoiding unnecessary or premature hedging during normal conditions.

Current decision processes often rely on high-level national storage indicators, expert judgement, and average price expectations. However, a large body of risk management literature shows that exposure in electricity markets is dominated by

infrequent but severe price events rather than by smooth movements in mean prices (Weron, 2014; Hull, 2018).

For organisations with wholesale exposure, delayed or poorly targeted hedging actions can therefore result in disproportionate cost impacts, while overly conservative hedging during benign conditions can increase costs without materially reducing risk. From a risk management perspective, the central challenge is not predicting the most likely price outcome, but identifying conditions under which downside risk escalates materially and persists over relevant planning horizons (Duffie & Singleton, 2003).

The business question can therefore be stated as follows: how can hydrological information be translated into reliable, forward-looking indicators of wholesale price risk that support timely and proportionate hedging decisions?

1.3 Importance and impact of the problem

From a governance perspective, the absence of transparent and analytically defensible risk signals can weaken confidence in decision-making and make it difficult to justify risk actions internally. As hydrological variability becomes more pronounced and dry-year risk remains a persistent feature of the New Zealand electricity system, improving the quality of hydro-risk monitoring is increasingly important.

A structured analytical framework that links hydrological conditions to price risk, particularly tail risk, has the potential to improve budget stability, strengthen risk governance, and enhance the credibility of hedging decisions. The impact of this project therefore extends beyond technical modelling to practical decision support within an applied organisational setting.

1.4 Expected organisational contribution

The expected contribution of this project to Wellington Electricity is the development of a regime-based hydro-risk framework that is both analytically sound and operationally usable. Rather than producing a single forecasting model, the project delivers a set of interpretable risk indicators that characterise how price risk changes across hydrological conditions.

Key contributions include:

- A structured classification of hydrological storage regimes that reflects underlying system stress rather than short-term price movements.
- Quantification of regime-conditioned price risk, with explicit emphasis on upper-tail outcomes relevant for risk management.

- A forward-looking, scenario-based framework that links regime persistence to accumulated extreme price exposure over planning horizons relevant to hedging decisions.
- A conceptual monitoring and dashboard design that translates complex analytical outputs into governance-relevant signals.

Collectively, these outputs support more timely and defensible risk decisions and provide a foundation for integrating hydrological risk into routine monitoring and review processes.

1.5 Key assumptions and limitations

Several assumptions and limitations shape the scope of the analysis. First, the project relies primarily on secondary, publicly available hydrological and market datasets. While these sources provide broad coverage and transparency, they constrain the level of operational detail that can be incorporated.

Second, the analysis focuses on hydrological drivers of price risk and does not attempt to model all short-term market influences, such as plant outages, fuel price shocks, or intraday demand fluctuations. These factors are acknowledged but treated as background noise relative to the long-horizon hydrological signal.

Third, storage regimes are defined using long-term historical data and are treated as stable over time. This helps ensure that changes in price risk are interpreted as responses to hydrological conditions rather than market outcomes. However, the framework may not fully capture future structural shifts in market behaviour.

Finally, the applied nature of the project requires a balance between analytical sophistication and interpretability. Some modelling choices prioritise governance relevance over maximum statistical complexity.

Chapter 2: Desk Research and Literature Review

This chapter establishes the structural and theoretical foundation for the empirical framework developed in Chapter 3. Section 2.1 outlines the organisational and market context from which the decision problem emerges. Section 2.2 reviews academic literature on hydro-dominated electricity markets and price risk. Sections 2.3 and 2.4 describe the data sources and institutional context supporting the analysis. Finally, Section 2.5 synthesises these strands to motivate the regime-based analytical design adopted in the subsequent methodology.

2.1 Organisation description and market context

Wellington Electricity operates as a regulated electricity distribution business within the New Zealand electricity sector. Although it does not participate directly in generation, it is exposed to wholesale electricity prices through procurement and hedging arrangements. Its cost outcomes are therefore influenced by wholesale market conditions, particularly during periods of elevated volatility.

The New Zealand electricity system is structurally characterised by a high reliance on hydro generation. Official energy statistics confirm that hydro generation remains the dominant source of electricity supply in New Zealand (MBIE, 2023). Unlike predominantly thermal systems, hydro-dominated markets depend on water storage as an intertemporal energy reserve. Storage levels determine the system's available flexibility and influence how generation is scheduled across time.

Because water stored today represents potential generation tomorrow, storage conditions function as a system-wide state variable. This institutional and physical structure creates a structural link between hydrological conditions and wholesale price risk. Understanding this linkage is central to evaluating how hydrological variability translates into cost exposure for market participants such as Wellington Electricity.

2.2 Academic literature informing the problem

Research on hydro-dominated electricity markets shows that water availability is a core driver of wholesale price behaviour. In hydro systems, storage works like a fuel inventory. Decisions to use water now versus save it for later mean that prices can move based on expected future water conditions, not only what is happening in the current month. This helps explain why wholesale prices can rise before storage reaches extreme lows, and why market outcomes can change quickly once the system begins to tighten. Similar intertemporal dynamics have been documented in hydro-dominated Nordic markets (Gjolberg & Johnsen, 2001; Fleten & Lemming, 2003; Lucia & Schwartz, 2002; Joskow, 2008; Joskow, 2019). For the New Zealand

context, this implies that national storage is not just a background indicator. It can be read as a system-level signal about how scarce flexible supply is likely to be.

The hydrology literature also highlights that dry conditions often persist across months rather than reverting immediately. Drought is typically a process, not a one-off shock. Once inflows weaken and storage declines, the system can stay stressed for an extended period, which matters because electricity risk is often managed over multi-month horizons (Tallaksen & Van Lanen, 2004; Mishra & Singh, 2010; Van Loon, 2015).

In parallel, electricity market studies consistently show that risk is driven by extreme price outcomes rather than average prices. A small number of high-price months can account for a large part of total cost exposure, especially in markets with tight capacity and limited short-run flexibility (Knittel & Roberts, 2005; Longstaff & Wang, 2004; Geman & Roncoroni, 2006).

This means a focus on average price movements can miss the main risk that matters for budgeting and hedging decisions (Weron, 2014; Mount et al., 2006).

However, a clear gap remains for applied decision support in a hydro-driven market setting like New Zealand. Much of the academic work either targets short-term price forecasting, or estimates complex structural or regime-switching models that are harder to translate into simple monitoring signals for practitioners. The literature is strong on explaining mechanisms, but it is less explicit about how to convert observable storage conditions into operational indicators that answer a decision-maker's question: when does risk start to escalate, and how likely is it to stay elevated long enough to create material tail exposure over a hedging horizon (Weron, 2014; Joskow, 2019).

This project addresses that gap by using an interpretable regime classification based on long-run storage data, then measuring wholesale price risk in distributional terms within each regime, with specific emphasis on extreme outcomes and persistence. The goal is not to produce a point forecast of prices, but to build a transparent link between storage states, persistence of stress, and tail-risk exposure that can be used in routine monitoring and hedging review.

2.3 Internal company materials and data sources

This study does not rely on proprietary internal Wellington Electricity datasets. Instead, it is based on publicly available regulatory and market data, ensuring transparency and reproducibility. Hydrological conditions are represented using naturalised inflow and storage information obtained from regulatory sources. Long-run inflow data, including the Mohaka naturalised flow series, span the period 1931–2024 and are used to contextualise dry-year risk and long-horizon hydrological

variability. Monthly storage data for major hydro lakes are available from 1980–2024 and are used to construct storage indicators and regime classifications.

Wholesale electricity prices are measured using monthly spot price data for the modern market period 2018–2024, yielding 84 monthly observations. Prices are expressed in nominal NZD/MWh. Given the applied objective of comparing relative risk across hydrological regimes and the short sample length, no inflation adjustment is applied.

2.4 Integration of external sources

In addition to the core hydrological and wholesale price datasets, selected external institutional sources are used to support interpretation of observed market behaviour. In particular, Transpower HVDC flow information is referenced to provide context on inter-island transmission conditions when interpreting price behaviour that may reflect transmission constraints or changes in inter-island energy transfer rather than hydrological conditions alone (Transpower New Zealand, 2023).

HVDC flow data are used in a targeted and descriptive way to triangulate interpretation of results, helping distinguish hydrology-linked stress from periods where price patterns may be influenced by transmission conditions. The HVDC information is drawn from publicly available Transpower system data and is used for contextual interpretation rather than as an analytical driver.

These external sources are not incorporated as direct inputs to regime construction or model estimation. Instead, they are used to ensure that empirical findings are discussed within the broader institutional and physical structure of the New Zealand electricity system, strengthening the credibility of interpretation without expanding the scope of the core hydro-risk framework.

2.5 Synthesis linking desk research to the analytical approach

Taken together, the literature shows that wholesale electricity risk in hydro-dominated systems is shaped less by short-term price movements and more by how storage conditions evolve and persist over time. Existing studies explain the mechanisms linking water availability to prices, but they offer limited guidance on how those insights can be converted into simple, forward-looking risk signals that support operational decisions such as hedging and cost control. For decision-makers, the key questions are not whether prices will increase in a specific month, but when risk begins to escalate, how severe that risk could become under adverse conditions, and whether elevated risk is likely to persist. This motivates an approach that classifies observable storage conditions into interpretable regimes and evaluates wholesale price risk in distributional terms within each regime, with particular attention to extreme outcomes and persistence.

Chapter 3: Research Methodology

3.1 Analytical design and time-scale alignment

Building on the literature synthesis in Chapter 2, this study adopts an applied analytical design that focuses on how observable hydrological conditions translate into decision-relevant wholesale electricity price risk. Rather than aiming to forecast prices at a fine temporal scale, the methodology is structured to capture when risk escalates, how severe it can become under adverse conditions, and whether elevated risk is likely to persist.

To support this objective, the analysis uses a staged time-scale alignment across its main components. Long-run hydrological and storage data are used to identify and define regimes that reflect sustained system stress, while wholesale electricity price risk is evaluated using a shorter, modern-market sample that reflects current market structure and regulatory conditions. This separation allows persistent hydrological signals to inform regime classification without imposing unrealistic data requirements on price-based risk analysis.

By aligning each component with the time horizon most relevant to its role, the framework provides a clear link between hydrological conditions and distributional measures of price risk, while maintaining consistent monthly alignment across the empirical analysis. This time-scale alignment across hydrology, storage, prices, and simulation ensures that the methodology remains interpretable and directly relevant for risk monitoring and hedging decisions.

Table 3.1: Time-scale alignment across analytical components.

Analytical component	Data source	Time period	Purpose
Long-run inflow baseline	Mohaka naturalised flow	1931 – 2024	Characterisation of long-term dry-year risk
Storage indicators	Major hydro lakes (aggregated)	1980 – 2024	Regime construction and transition modelling
Wholesale electricity prices	Monthly spot prices	2018 – 2024 (84 months)	Regime-conditioned price risk analysis
Regime-based simulation	Regime transitions + price distributions	Conditional on regimes	Scenario-based risk assessment

Note: This table summarises the data sources, time periods, and analytical roles of each component within the staged analytical framework. Long-horizon hydrological data inform regime construction, while price-based risk analysis is conducted using a shorter, policy-relevant market sample.

3.2 Data sources

Hydrological conditions are represented using naturalised inflow and storage information obtained from publicly available regulatory sources. Long-run inflow information, including the Mohaka naturalised flow series spanning 1931–2024, is used to characterise dry-year risk and to provide historical context for long-term hydrological variability (Tallaksen & Van Lanen, 2004; Mishra & Singh, 2010; Van Loon, 2015).

Storage information from major hydro lakes is available on a monthly basis from 1980–2024 and is used to construct storage indicators and regime classifications. These data reflect the physical availability of hydro energy and form the basis for defining sustained system states.

Wholesale electricity prices are measured using monthly spot price data for the modern market period 2018–2024, yielding a total of 84 monthly observations. Prices are expressed in nominal NZD/MWh. Given the applied objective of comparing relative risk across hydrological regimes and the limited sample length, no inflation adjustment is applied (Weron, 2014; Joskow, 2019).

In addition, high-frequency national load data for January 2026 are used in Appendix A for supplementary descriptive analysis. These data are not incorporated into the core modelling framework.

3.3 Data preparation and storage indicators

Prior to analysis, hydrological, storage, and price data are organised on a common monthly time index. Temporal alignment ensures that observed market outcomes correspond to contemporaneous physical system states.

Storage indicators are constructed by aggregating lake-level storage observations within relevant hydro systems and are further aggregated to a national-level indicator representing overall system water availability. These indicators are derived solely from physical storage information and do not incorporate any price-related variables.

Importantly, storage indicators and regime classifications are constructed independently of wholesale price information. This separation ensures that regime definitions reflect underlying physical system conditions rather than short-sample market outcomes, allowing price behaviour to be analysed conditionally on exogenously defined hydrological regimes.

3.4 Construction of hydrological storage regimes

Monthly hydrological storage regimes are constructed using a national storage index derived from aggregated hydro lake storage levels. Storage states are discretised into three regimes: High, Medium, and Low.

Regime thresholds are calibrated using the empirical distribution of the storage index over the full 1980–2024 sample period and are fixed ex ante for all subsequent analysis. The Medium regime represents typical operating conditions, while the High and Low regimes capture surplus and stress states respectively.

This discretisation translates continuous physical storage information into a small number of economically interpretable system states that can condition price risk and support decision-making.

3.5 Regime dynamics and transition modelling

To capture the temporal evolution of system states, regime dynamics are modelled using a first-order Markov transition framework (Hamilton, 1989; Gray, 1996; De Jong, 2006). Monthly regime transitions are identified by comparing each month's regime state with that of the preceding month over the full storage sample.

Transition counts are aggregated and converted into conditional transition probabilities, providing a descriptive representation of regime persistence and transition structure. This transition framework supplies key inputs for scenario construction and forward-looking risk assessment in later stages of the analysis.

3.6 Regime-conditioned price risk framework

Wholesale electricity price risk is assessed conditionally on hydrological storage regimes rather than through an unconditional or point-forecasting approach. Monthly price outcomes are aligned with contemporaneous regime states, allowing price behaviour to be examined separately under normal and stressed system conditions.

Risk characterisation focuses on tail behaviour rather than average prices, consistent with structural models of electricity price spikes and jump dynamics (Kanamura & Ōhashi, 2007; Geman & Roncoroni, 2006; Mount et al., 2006). Regime-specific exceedance probabilities and empirical price distributions are used to quantify the likelihood and severity of adverse outcomes under different hydrological states.

This design reflects the applied risk-management context of electricity markets, where infrequent but severe price outcomes can dominate cost exposure.

3.7 Simulation design and monitoring concepts

To extend regime-conditioned risk characterisation into forward-looking decision support, scenario-based simulations are constructed by combining empirical regime transition dynamics with regime-conditioned price distributions.

The simulation framework generates distributions of potential price outcomes over fixed planning horizons under alternative hydrological stress scenarios. Simulation outputs support the construction of applied risk metrics such as the probability of at least one extreme price event within the horizon, distributions of maximum prices, and upper-tail quantiles.

Monitoring concepts are defined as functions of regime state, regime-entry risk, and regime persistence. These concepts are designed to translate analytical outputs into interpretable indicators suitable for risk monitoring and hedging review, without reliance on precise point forecasts, forming part of the overall staged analytical workflow.

Table 3.7: Overview of the analytical workflow.

Stage	Input	Methodological component	Output (method-level)
1	Storage and inflow data	Construction of national storage indicators	Monthly storage indicators
2	Storage indicators	Regime classification	High / Medium / Low regimes
3	Regime sequence	Markov transition modelling	Transition probability matrix
4	Prices + regimes	Regime-conditioned risk framework	Regime-conditioned tail risk metrics
5	Regimes + prices	Scenario-based simulation design	Planning-horizon exposure metrics

Note: This table outlines the sequential structure of the analytical framework, highlighting how data inputs are transformed through methodological components into intermediate, method-level outputs that support subsequent analysis.

3.8 Alternative approaches, tools, and limitations

Several alternative analytical approaches were considered during the design of the study. Unconditional price models were not adopted due to their inability to capture state-dependent risk asymmetry. More complex structural or multivariate models were also considered but rejected due to data availability constraints and the applied objective of producing interpretable, governance-relevant indicators.

Data preparation and exploratory analysis are conducted using SQL, while modelling and simulation are implemented in Python within a Jupyter Notebook environment to support transparency and reproducibility. The framework prioritises interpretability and decision relevance over maximum statistical complexity, consistent with applied risk-management design principles that emphasise structured monitoring and downside protection (Jarrow & Turnbull, 2000; Hull, 2018).

3.9 Ethical considerations

This study relies exclusively on secondary, publicly available datasets obtained from regulatory and market sources. No personal, confidential, or commercially sensitive data are used. All analysis is conducted at an aggregated system level, ensuring minimal ethical and privacy risk.

Chapter 4: Data Analysis

This chapter presents the empirical analysis linking hydrological storage conditions to wholesale electricity price risk in New Zealand. The objective is not to forecast short-term prices, but to examine how observable storage states translate into state-dependent risk and how that risk accumulates over decision-relevant horizons.

The empirical analysis follows the staged workflow outlined in Chapter 3. After constructing storage regimes and examining their transition dynamics, wholesale price risk is evaluated conditionally on regime states, with emphasis on tail behaviour. Early-warning indicators based on regime-entry risk and persistence are then assessed. Finally, regime transition dynamics and regime-conditioned price distributions are integrated within a forward-looking simulation framework to quantify planning-horizon exposure.

Results are organised to highlight the main empirical findings that directly inform managerial interpretation.

4.1 Data coverage and analytical context

The empirical analysis combines long-horizon hydrological information with a shorter modern-market price sample to ensure both statistical grounding and decision relevance. Storage regimes and regime transition dynamics are constructed from monthly hydro lake storage data over 1980–2024, while wholesale electricity price risk is evaluated using monthly spot prices from 2018–2024. All series are aligned at a monthly frequency to maintain consistency across regime identification, price conditioning, early-warning evaluation, and simulation.

Long-run naturalised inflow information (1931–2024) is used to characterise historical dry-year severity and to define the dry-year scenario labels applied in the simulation stage, rather than as a direct input to regime estimation. This preserves the staged design described in Chapter 3: structural hydrological conditions determine regime states and persistence, while price distributions are estimated from the policy-relevant modern market period.

Table 4.1 summarises the datasets used in the empirical chapter and their roles. The separation between long-horizon hydrological structure and the modern price sample ensures that persistence and transition behaviour are estimated using robust physical records, while risk measures and exposure metrics remain anchored to the market environment in which hedging decisions are made.

Table 4.1: Key datasets used in the empirical analysis.

Dataset	Frequency	Period	Analytical role
National hydro storage	Monthly	1980 – 2024	Regime definition, persistence
Basin storage (Taupō, Hāwea)	Monthly	1980 – 2024	Early warning indicators
Wholesale prices	Monthly	2018 – 2024	Regime-conditioned price risk

Notes: National storage indicators are constructed from aggregated hydro storage data. Basin-level indicators are based on storage observations for Lake Taupō and Lake Hāwea. Wholesale electricity prices are expressed in NZD/MWh.

Section summary:
The data design separates structural hydrological dynamics from modern price behaviour, allowing regime-based risk analysis without imposing unrealistic historical price requirements. This separation supports both interpretability and decision relevance.

4.2 Storage regime dynamics and persistence

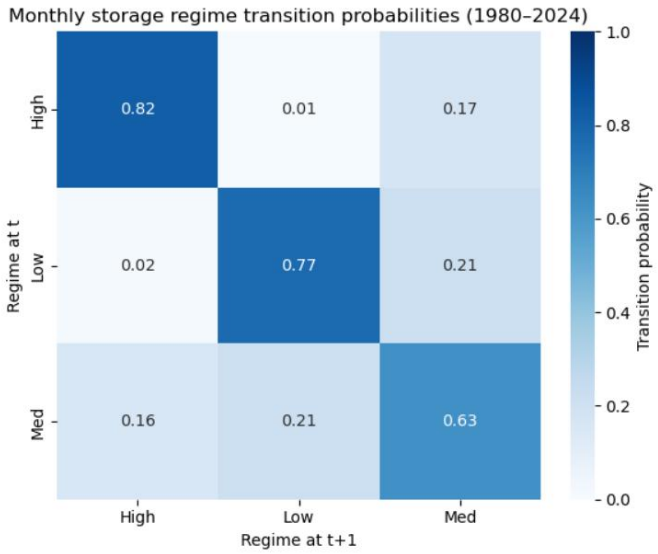
National hydrological storage regimes exhibit substantial temporal persistence. Estimated transition probabilities indicate a high likelihood of remaining in the same regime from one month to the next, particularly for High- and Low-storage states. Direct transitions between High and Low regimes are rare and typically occur through the Medium regime, indicating gradual evolution rather than abrupt shifts.

Transition matrix estimates show that the probability of remaining in the High regime exceeds 0.8, while persistence in the Low regime exceeds 0.75. Direct High-to-Low transitions account for only a small proportion of movements.

This persistence has structural implications for risk. If regime states were short-lived, monthly exceedance probabilities would not meaningfully accumulate. However, when stressed regimes persist across multiple months, even moderate conditional probabilities of extreme prices compound over a planning horizon. In effect, persistence introduces a duration channel through which hydrological stress translates into cumulative exposure.

Moreover, the asymmetry between persistence and transition implies that risk escalation is gradual and path-dependent. The system typically moves from High to Medium before reaching Low. This creates an observable deterioration pathway rather than a discontinuous jump, supporting the feasibility of structured monitoring.

Figure 4.2: Monthly storage regime transition probabilities.



Notes: Rows denote the current-month regime and columns denote the subsequent-month regime. Cell values report conditional transition probabilities.

Section summary:
Hydrological stress is persistent rather than transient. Persistence transforms monthly risk into multi-month exposure and makes cumulative tail risk structurally plausible. This finding supports monitoring designs that account for duration, not only state.

4.3 Regime-conditioned wholesale electricity price distributions

Wholesale electricity price behaviour differs systematically across hydrological storage regimes. Differences are most pronounced in the upper tail of the distribution rather than in average prices.

Exceedance probabilities for high-price thresholds increase steadily as storage conditions deteriorate. The probability of observing a monthly price above NZD 200/MWh rises from 0.040 in High regimes to 0.132 in Low regimes. This pattern indicates strong state dependence in extreme price risk.

Table 4.3: Probability of monthly wholesale electricity prices exceeding a high-price threshold.

Storage Regime	Number of months	Pr(price > 200 NZD/MWh)
HIgh	25	0.040
Med	21	0.095
Low	38	0.132

Notes: Probabilities are computed using monthly average wholesale prices. Regime classification is based on contemporaneous national storage conditions.

Distributional analysis reinforces this pattern. Regime-conditioned empirical cumulative distribution functions (ECDFs) and survival functions show increasing divergence in the right tail as storage deteriorates. While median price differences are moderate, the separation widens materially at higher quantiles.

This implies that hydrological regimes primarily affect the shape of the price distribution rather than shifting it uniformly. In risk terms, the impact of storage deterioration is convex: tail outcomes expand disproportionately relative to changes in central tendency. Such convexity reflects nonlinear amplification mechanisms often observed in financial time series (Hsieh, 1991) and is consistent with scarcity pricing mechanisms, where tight supply conditions amplify extreme outcomes more than average outcomes.

From a decision perspective, this distinction matters. Hedging value is concentrated in protecting against infrequent but severe outcomes. Focusing only on mean prices would understate the primary exposure channel associated with Low-storage regimes.

Figure 4.3A: Regime-conditioned ECDFs of wholesale electricity prices.

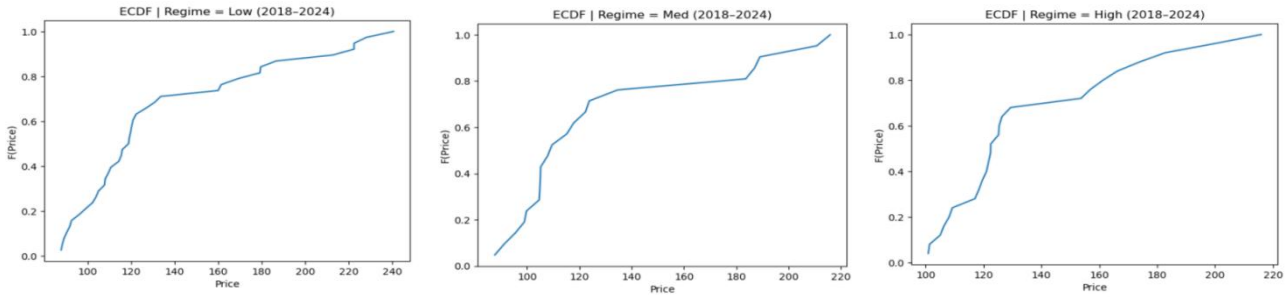
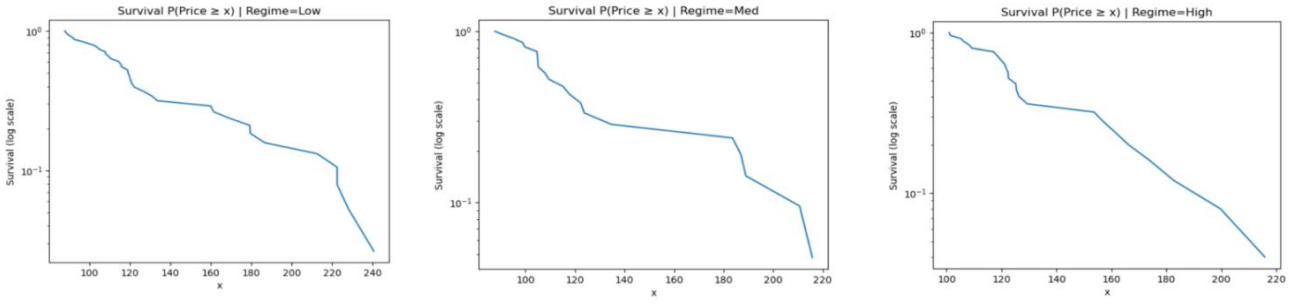


Figure 4.3B: Regime-conditioned survival functions of wholesale electricity prices.



Notes: Prices are monthly averages expressed in NZD/MWh. Survival functions are shown on a log scale to highlight tail behaviour.

Section summary:
Wholesale price risk is strongly state-dependent. Storage deterioration disproportionately expands upper-tail outcomes rather than uniformly raising prices. Tail risk, not average price change, is the primary exposure channel.

4.4 Regime entry risk and basin-level early warning indicators

Transitions into Low-storage regimes are not random and can be partially anticipated using basin-level storage indicators. Basin-standardised storage measures for Lake Taupō and Lake Hāwea show systematic deterioration prior to national regime reclassification.

Months classified as "Future Low Regime = Yes" exhibit materially lower mean and median basin z-scores relative to non-entry months. This suggests that local basin stress precedes national regime deterioration.

Table 4.4: Basin-level standardised storage indicators conditional on subsequent entry into a Low-storage regime.

Basin	Future Low Regime	Mean	Median	Std	N
Lake Taupō	No (0)	0.47	0.65	0.84	284
Lake Taupō	Yes (1)	-0.53	-0.53	0.89	256
Lake Hāwea	No (0)	0.29	0.11	1.16	284
Lake Hāwea	Yes (1)	-0.32	-0.31	0.65	256

Notes: Basin indicators are monthly z-scores. "Future Low Regime" indicates months followed by entry into a Low-storage regime within a three-month forward window. N denotes the number of monthly observations.

Mechanistically, this reflects the uneven development of hydrological stress across regions. Stress often emerges in individual basins before being reflected in aggregated national storage measures. Basin indicators therefore provide additional probabilistic information about impending regime shifts.

Importantly, these signals are probabilistic rather than deterministic. The objective is not precise prediction of regime change dates, but structured escalation when transition probability rises materially above baseline.

To evaluate whether these signals are usable under an ex ante monitoring design rather than only descriptively informative, a rolling out-of-sample logistic framework was implemented using the same three-month forward entry definition. At each evaluation point, model estimation relied solely on information available at that time, ensuring that predictive performance reflects genuine real-time capability under an expanding-window design.

Using only the national storage index, the rolling evaluation yields an out-of-sample AUC of 0.759 and a Brier score of 0.209. These results indicate meaningful discrimination in predicting entry into Low-storage regimes despite the inherently stochastic nature of hydrological transitions.

When basin-level storage indicators for Taupō and Hāwea are incorporated, in-sample discrimination improves from 0.830 to 0.841. Although incremental in magnitude, this improvement occurs in a setting where baseline predictability is already high, indicating that basin-level conditions contain structural information not captured by aggregate national storage alone.

Coefficient estimates are economically and physically coherent. Lower storage levels increase the probability of subsequent entry into a Low regime, with Taupō exhibiting the largest magnitude effect, followed by Hāwea, and national storage remaining statistically significant but comparatively less sensitive. This ranking supports the interpretation that national regime deterioration reflects the accumulation of basin-specific storage deficits rather than purely aggregate fluctuations.

Taken together, these findings demonstrate that regime transitions are not purely aggregate phenomena but reflect the build-up of identifiable basin-level stress. Incorporating basin indicators enhances both the interpretability and operational relevance of regime-based monitoring frameworks.

Section summary:
Basin-level indicators contain forward-looking information about regime deterioration. Rolling out-of-sample validation confirms that these signals retain predictive power under an ex ante design. Early-warning value lies in probabilistic escalation signals rather than precise timing forecasts.

4.5 Forward-looking simulation of extreme price risk and cost exposure

To quantify forward-looking exposure, regime-conditioned price distributions are combined with estimated regime transition dynamics within a Monte Carlo framework. Regime evolution follows the transition probabilities estimated in Section 3.5, and regime-specific price outcomes are generated using gamma distributions calibrated to regime-level empirical price data, capturing the right-skewed and strictly positive nature of wholesale electricity prices consistent with observed market behaviour.

This structure explicitly links three components:

- Current regime state
- Transition and persistence dynamics
- Regime-specific price distribution

By simulating joint regime paths and price realisations over a 12-month horizon, the framework captures the interaction between duration and tail behaviour.

Simulated price paths reveal that sustained Low-storage conditions materially increase the probability of at least one extreme price month within the planning horizon. Mean annual price differences are moderate, but the distribution of maximum annual prices exhibits pronounced right-tail expansion.

Figure 4.5: Simulated distribution of maximum wholesale electricity prices.



Notes: Simulations combine regime-conditioned gamma distributions with estimated Markov transition probabilities over a 12-month planning horizon. Results are based on repeated Monte Carlo paths.

To translate distributional risk into economic exposure, simulated prices are converted into annual wholesale cost per MW of load. Results distinguish expected cost from upper-tail (P95) exposure.

Table 4.5: Expected and P95 wholesale cost exposure under hydrological scenarios.

Scenario	Expected Cost (NZ$ / MW-year)	P95 Cost (NZ$ / MW-year)	Δ Expected Cost vs Non-dry	Δ P95 Cost vs Non-dry
Non-dry (10y)	2,304,415	2,870,251	0	0
Dry10 (10y)	2,326,609	2,901,575	22,194	31,324
Dry20 (10y)	2,332,230	2,911,158	27,815	40,906

Notes: Costs are expressed in NZD per MW-year and are computed from simulated monthly price paths assuming a constant 1 MW annual load. P95 denotes the 95th percentile of simulated annual cost outcomes. Incremental values are relative to the non-dry baseline.

The divergence between expected cost and P95 cost under dry scenarios illustrates the non-linear exposure channel. While average cost increases are modest, worst-case exposure expands more sharply. This confirms that hydrological stress primarily affects downside risk rather than central expectation.

From a governance perspective, this provides a quantitative basis for linking storage regimes to hedging discussions framed around risk appetite and downside protection rather than average procurement cost.

Section summary:
Regime persistence and tail expansion jointly generate material planning-horizon exposure. Simulation confirms that hydrological stress disproportionately increases downside risk, validating a tail-focused monitoring approach.

4.6 Chapter summary

The empirical analysis establishes four core findings:

- Storage regimes are persistent and evolve gradually
- Wholesale price risk is state-dependent, with deterioration primarily affecting upper-tail outcomes
- Basin indicators provide early probabilistic signals of regime deterioration
- Persistence and tail behaviour interact to produce non-linear exposure over planning horizons

This analytical foundation supports the decision-focused interpretation developed in the subsequent discussion chapter. Supplementary high-frequency descriptive evidence is provided in Appendix to contextualise tail-risk behaviour but is not incorporated into the core modelling framework.

Chapter 5: Discussion

5.1 Linking findings to the business decision problem

Chapter 1 defined the central business decision problem for Wellington Electricity: how to improve the timing and structure of hedging responses under hydrological uncertainty. The objective was not to forecast monthly prices precisely, but to translate storage information into forward-looking, structured risk signals that support disciplined escalation when risk materially increases.

The findings in Chapter 4 provide a direct answer:

First, hydrological storage regimes are persistent rather than short-lived. Once the system enters a Low-storage state, the probability of remaining under stress in subsequent months rises materially. This confirms that storage conditions shape the risk environment over the same multi-month horizons in which procurement and hedging decisions are reviewed. Regime persistence therefore acts as a structural mechanism through which hydrological stress accumulates into sustained exposure over the hedging horizon.

Second, regime deterioration affects the shape of the price distribution more than its mean. Under Low-storage regimes, the probability of extreme price outcomes increases significantly, while average price changes remain comparatively moderate. For a retailer exposed to wholesale markets, this distributional shift is economically more important than small movements in expected prices. Budget risk is driven by clustering of extreme months, not gradual mean increases.

Third, storage-based early-warning indicators retain predictive power under a strict rolling out-of-sample design. National storage alone provides meaningful discrimination in predicting entry into a Low-storage regime within a three-month forward window. Basin-level indicators add incremental information beyond national aggregates, especially when regional deterioration precedes national reclassification.

Together, these findings demonstrate that hydrological information can be translated into forward-looking, probabilistic risk signals suitable for governance use. The framework therefore addresses the original decision problem defined in Chapter 1: improving the timing and structure of hedge review under hydrological stress.

5.2 Interpretation in operational terms

The empirical results imply a shift in how storage conditions should be interpreted in practice.

Storage is not merely a contemporaneous system snapshot. Because regimes exhibit persistence, current storage conditions contain information about the likely duration of stress. Risk should therefore be evaluated over a planning horizon rather than on a single-month basis.

Operationally, three implications follow:

First, monitoring should focus on regime escalation and persistence rather than short-term price fluctuations. Temporary price movements in a High or Medium regime do not carry the same governance implications as equivalent movements during Low-storage stress.

Second, risk discussions should explicitly incorporate exceedance probabilities and downside exposure metrics. Since hydrological deterioration disproportionately expands the upper tail of the distribution, average price expectations alone understate material exposure.

Third, hedging review timing should align with rising regime-entry probabilities rather than waiting for realised price spikes. The out-of-sample validation confirms that storage-based models provide usable ex ante signals. Basin indicators strengthen these signals when regional stress builds before national reclassification.

This interpretation transforms storage monitoring from descriptive reporting into structured risk escalation management.

5.3 Relationship to existing literature

Chapter 2 identified three structural themes in the literature: hydro systems function like inventory markets; drought conditions tend to persist; and electricity market risk is dominated by extreme outcomes rather than smooth mean shifts.

The empirical findings are consistent with these themes and confirm their relevance in the New Zealand context.

Regime persistence aligns with hydrological research emphasising duration effects in drought processes. The concentration of risk in the upper tail reflects established evidence that a small number of extreme events often account for a disproportionate share of annual cost exposure. The incremental predictive value of basin indicators supports the view that regional hydrological dynamics can contain early structural signals not immediately visible in aggregated national measures.

However, the contribution of this project is not a novel forecasting algorithm. Its value lies in integrating these structural insights into a transparent monitoring chain that connects physical storage dynamics to governance decisions. Rather than maximising short-horizon price forecast accuracy, the framework prioritises

interpretability, stability, and alignment with managerial risk questions. This directly addresses the practical gap identified in Chapter 2: converting hydrological structure into repeatable and defensible decision triggers.

5.4 Monitoring triggers and decision framework

The results support the use of regime-aware monitoring triggers rather than reactive responses to individual price movements.

A practical escalation rule can be defined around forward-looking regime-entry probability. For example, when the predicted probability of entering a Low-storage regime within the next three months exceeds a predefined threshold, a formal hedge review is initiated.

This trigger does not mandate automatic execution. Instead, it formalises escalation discipline. The objective is to ensure that rising hydrological stress is addressed consistently and transparently, reducing reliance on subjective judgement or delayed reaction.

Basin-level indicators function as an escalation layer. When Taupō or Hāwea storage shows systematic deterioration ahead of national regime change, monitoring intensity can increase even before aggregate indicators cross regime boundaries.

This trigger logic reframes hedging discussions around probabilistic stress escalation and downside exposure rather than short-term price noise. It links storage structure directly to governance timing.

5.5 Dashboard integration

To operationalise the analytical framework, outputs must be consolidated into a structured monitoring dashboard.

A regime-aware dashboard should integrate four components:

- Current regime classification and persistence metrics
- Regime-conditioned tail risk indicators
- Forward-looking probability of Low-regime entry
- Estimated cost exposure over the planning horizon

Table 5.5: Integration of regime signals into monitoring dashboard.

Dashboard Module	Key Indicator	Analytical Source	Decision Use
Storage regime state	Current regime classification; persistence probability	Section 4.2	Assess current stress and likelihood of continuation
Forward-looking regime risk	Probability of Low-regime entry (3-month horizon)	Section 4.4	Trigger structured hedge review when escalation risk rises
Price tail risk	Regime-conditioned exceedance probability	Section 4.3	Evaluate downside exposure beyond mean price changes
Cost exposure	Expected cost and P95 cost estimate	Section 4.5	Translate regime stress into financial impact

Presenting these signals jointly reduces overreaction to temporary market movements and strengthens consistency in procurement discussions. The dashboard serves as the delivery mechanism for disciplined risk review rather than an additional analytical layer.

5.6 Limitations

Several limitations should be acknowledged：

First, the wholesale price sample is relatively short, limiting the number of observed extreme price episodes. Tail probability estimates therefore remain sample-sensitive. While regime conditioning enhances interpretability, uncertainty around very rare outcomes persists.

Second, the framework focuses intentionally on hydrological drivers and does not explicitly model other risk factors such as generation outages, fuel shocks, or demand anomalies. The objective is structural monitoring rather than comprehensive short-term forecasting.

Third, probability-based triggers support disciplined review but do not replace managerial judgement. Contract structure, liquidity conditions, and existing hedge positions must still inform execution decisions.

Finally, regime definitions are assumed stable across the sample period. Structural changes in market design or generation mix may require periodic recalibration to maintain relevance.

These limitations do not invalidate the framework but define the conditions under which it should be interpreted and applied.

Chapter 6: Conclusion

6.1 Answers to the business decision problem

This project set out to address a practical decision problem for Wellington Electricity: how to translate hydrological information into forward-looking and defensible indicators of wholesale price risk that support timely hedging review, without relying on short-term price forecasting.

The central conclusion is that hydrological variability should be treated not as background market noise, but as a structured risk state that governs downside exposure. National storage conditions contain systematic information about how risk evolves, but the relevant signal is distributional rather than mean-based. Deteriorating storage modestly increases expected prices, but the dominant impact is on downside risk through right-tail expansion.

This distinction resolves the core decision tension identified in Chapter 1. The objective is not to predict next month's price, but to identify when exposure to rare but severe cost events becomes materially more likely and persistent. Because hydrological stress tends to endure once established, risk is inherently a horizon problem. Escalation is meaningful not because a single month is expensive, but because stressed conditions increase the likelihood that adverse outcomes accumulate over time.

The evidence further shows that regime deterioration can be identified probabilistically before full stress is realised. Early signals do not eliminate uncertainty, but they allow escalation to be framed in structured terms rather than as a reaction to realised price spikes. This supports a disciplined review process in which rising hydrological stress triggers formal evaluation of hedge positions before tail exposure fully materialises.

Finally, translating regime-conditioned risk into cost exposure clarifies why hedging value is concentrated in protecting the upper tail rather than stabilising average prices. The framework therefore answers the original business question directly: hydrological information can be converted into structured state indicators, escalation probabilities, and planning-horizon exposure metrics that support proportionate and transparent hedging decisions.

6.2 Contribution and implications for decision support

The contribution of this project is conceptual and organisational rather than algorithmic. It demonstrates that hydrological variability can be reframed from a descriptive background condition into a structured state variable for governance.

Instead of treating hydro conditions as contextual information reviewed alongside market outcomes, the framework positions storage as the conditioning variable through which risk is interpreted. This shift changes the logic of monitoring. Risk discussion becomes anchored in system state and persistence rather than in realised price movements.

The practical implication is not improved short-term prediction, but improved escalation discipline. By distinguishing between temporary price noise and structurally elevated stress, the framework supports proportionate and transparent review of hedge positions. It formalises when risk warrants attention, while avoiding overreaction during benign conditions.

Importantly, the framework does not attempt to eliminate uncertainty. Wholesale electricity markets remain inherently volatile and influenced by multiple drivers. Its value lies in governance coherence: aligning physical system stress with financial risk assessment over the same planning horizon.

Within its defined scope, the project establishes a defensible bridge between hydrological structure and risk oversight. In doing so, it strengthens the consistency, clarity, and accountability of wholesale risk management under uncertainty.

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Appendix

Supplementary High-Frequency Load Analysis

This appendix reports a supplementary high-frequency analysis of national electricity load dynamics that is not incorporated into the core price-risk modelling framework. The purpose of this analysis is exploratory and illustrative: to examine the characteristics of rare intra-day demand stress events and to clarify how short-term demand-side volatility may interact with tail-risk exposure in a hydro-dominated electricity system.

A.1. Data and aggregation

To complement the core hydro-storage and price modelling framework, this section introduces a high-frequency supplementary analysis of national electricity load dynamics. Using 5-minute interval national load data for the period **5–18 January 2026**, we identify and characterise extreme intra-day load spike events and contrast their volatility profiles against normal operating days.

National load data (NZ TOTAL, MW) were sampled at 5-minute resolution and aggregated to daily metrics including:

Daily mean load (MW)
Daily peak load (MW)
Daily trough load (MW)
Intra-day load range (peak − trough, MW)

A.2 Spike identification rule

To ensure a conservative and interpretable classification, a day is defined as a **spike day** if it satisfies both conditions:

Peak-to-mean ratio > **1.5**
Peak excess (peak − mean) > **1500 MW**

This dual-threshold rule avoids misclassifying structurally high-load days as spike events and focuses on genuine transient stress episodes.

Table A.2: Daily spike diagnostics (January 2026).

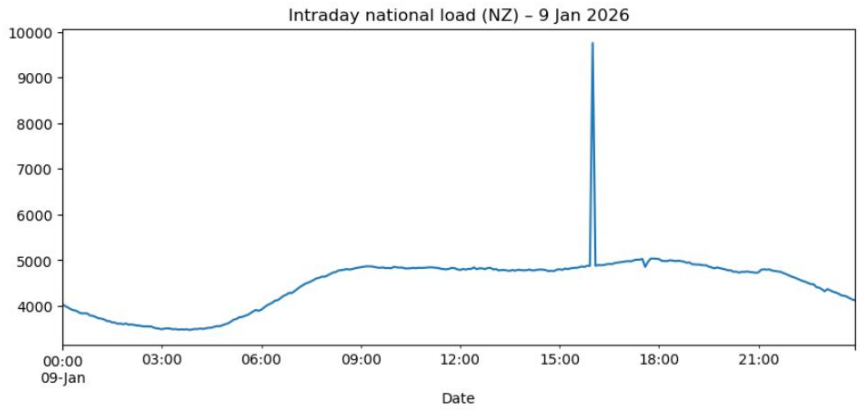
Date	Mean load (MW)	Peak load (MW)	Trough load (MW)	Spike ratio	Spike excess (MW)	Spike classification
2026-01-05	4053.46	4706.39	3022.10	1.16	652.93	Normal

2026-01-06	4209.13	4847.40	3213.43	1.15	638.27	Normal
2026-01-07	4345.11	5001.95	3331.71	1.15	656.84	Normal
2026-01-08	4427.54	5068.16	3441.92	1.14	640.62	Normal
2026-01-09	4472.14	9752.61	3466.19	2.18	5280.47	Extreme
2026-01-10	4277.51	4889.36	3469.53	1.14	611.85	Normal
2026-01-11	4219.33	4989.48	3218.56	1.18	770.15	Normal
2026-01-12	4514.03	5079.61	3430.80	1.13	565.58	Normal
2026-01-13	4500.49	5126.49	3455.82	1.14	626.00	Normal
2026-01-14	4553.86	5158.29	3485.07	1.13	604.43	Normal
2026-01-15	4606.01	5233.55	3435.50	1.14	627.54	Normal
2026-01-16	4402.20	4978.98	3356.95	1.13	576.78	Normal
2026-01-17	3998.82	4460.66	3237.05	1.12	461.84	Normal
2026-01-18	3797.06	6332.20	3109.28	1.67	2535.14	Moderate

A.3 Intra-day structure of spike events

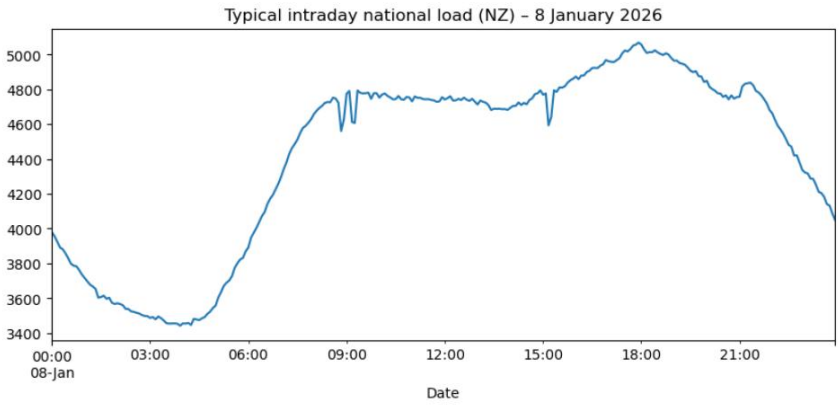
Visual inspection of the 9 January 2026 event confirms that the extreme peak is driven by a single, short-lived spike rather than a sustained elevation in load. Load levels immediately before and after the spike revert to a stable intra-day profile comparable to normal operating days.

Figure A.3-1: Intraday national load profile on spike day.



By contrast, non-spike days, such as 8 January 2026, exhibit smooth diurnal load patterns with moderate morning and evening ramps.

Figure A.3-2: Typical intraday national load profile on non-spike day.



A.4 Volatility comparison: spike vs non-spike days

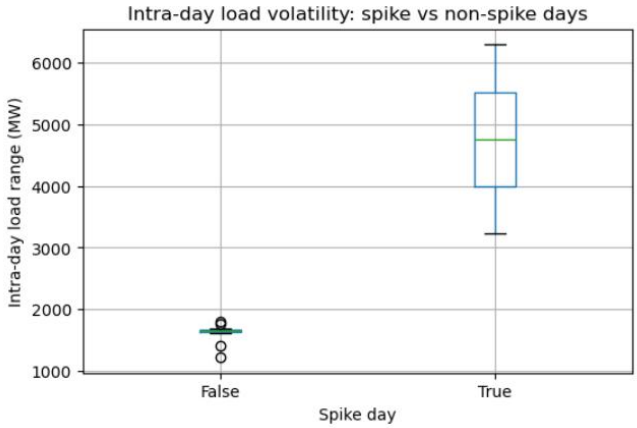
To quantify the impact of spike events on short-term system volatility, intra-day load ranges were compared across spike and non-spike days.

Table A.4: Intra-day load range (MW): spike vs non-spike days.

Classification	Mean	Median	Max	Count
Non-spike days	1620	1659	1798	12
Spike days	4755	4755	6286	2

Spike days exhibit intra-day load ranges nearly three times larger than normal days, highlighting the disproportionate contribution of rare extreme events to short-term system stress.

Figure A.4: Intra-day load volatility: spike vs non-spike days.



A.5 Implications for future extensions

The high-frequency load analysis provides contextual evidence that complements the main regime-based framework.

First, the identification of extreme intra-day spikes shows that short-term operational stress is typically concentrated in rare and transient events rather than sustained demand shifts. This supports the emphasis on tail behaviour in the core analysis.

Second, spike days exhibit substantially larger intra-day ranges than normal days, indicating that short-term volatility is highly asymmetric. Although infrequent, these events dominate volatility metrics, reinforcing the focus on downside exposure rather than average conditions.

Third, the clear distinction between spike and non-spike days suggests that event-based classification may serve as a useful extension to continuous modelling approaches. This aligns with the conditional strategy adopted in the main chapters, where system behaviour is evaluated differently across regimes.

Overall, these exploratory results strengthen the conceptual linkage between short-term operational stress and longer-horizon hydro-storage risk, without altering the structure of the core analytical framework.